

Stellar Trace: Host Profiling & Progenitor Physics

Kahoot 1-Page AI Import Template (30 Questions)

Q1: In a flat Lambda-CDM universe, which energy component dominates late-time cosmic expansion, and what is its dynamical effect?

A) Matter density, leading to gravitational recollapse (a Big Crunch) | **B)** Dark Energy (cosmological constant), leading to accelerated expansion | **C)** Radiation density, leading to a decelerating expansion rate | **D)** Spatial curvature, maintaining a static state

Correct: B

Q2: A photon is emitted from a galaxy at redshift $z = 1.0$. By what factor is its observed wavelength stretched compared to its rest-frame wavelength?

A) Stretched by a factor of 1.5 | **B)** Stretched by a factor of 2.0 | **C)** Stretched by a factor of 1.0 (no change) | **D)** Stretched by a factor of 0.5 (compressed)

Correct: B

Q3: When generating a simulated isotropic galaxy distribution, why does sampling the declination angle (Dec) uniformly in degrees from -90 to +90 fail to produce a uniform spatial density?

A) It causes galaxies to crowd unnaturally near the celestial poles (Dec = +/-90 degrees) | **B)** It causes galaxies to cluster along the celestial equator (Dec = 0 degrees) | **C)** It creates an artificial depletion at the polar coordinate singularities | **D)** It introduces a radial bias relative to the observer's location

Correct: A

Q4: Which physical mechanism is primarily responsible for the rapid quenching of star formation in massive galaxies ($M_{\text{star}} > M_c$), creating the exponential cutoff in the Stellar Mass Function?

A) Supernova feedback | **B)** Active Galactic Nuclei (AGN) feedback | **C)** Stellar wind stripping | **D)** Dark matter halo evaporation

Correct: B

Q5: Why is the faint-end slope ($\alpha = -1.25$) of the Stellar Mass Function relatively flat compared to the dark matter halo mass function?

A) Dwarf galaxies are dominated by active supermassive black holes | **B)** Supernova feedback and UV heating eject gas from shallow potential wells | **C)** Dwarf galaxies contain no dark matter to bind their gas | **D)** Cosmic expansion prevents accretion in low-mass halos

Correct: B

Q6: To generate mock galaxy masses following a Schechter mass function using a uniform random variable U in $[0,1]$, which mathematical technique is applied?

A) Pass U through the Inverse Cumulative Distribution Function (Inverse CDF) | **B)** Multiply U by the Hubble constant (H_0) | **C)** Convolve U with a Gaussian measurement error | **D)** Divide U by the critical density of the universe

Correct: A

Q7: In Inverse Transform Sampling, drawing a random value $U = 0.50$ corresponds to returning which parameter of the target stellar mass distribution?

A) The maximum stellar mass on the grid | **B)** The mean stellar mass of the distribution | **C)** The median stellar mass (50th percentile) | **D)** The mode (most common mass) of the distribution

Correct: C

Q8: Why must a stellar mass function, $\Phi(M)$, be multiplied by $\ln(10) \cdot M$ when evaluating it in logarithmic bins of $x = \log_{10}(M)$?

A) To account for stellar mass loss over cosmic time | **B)** To conserve probability density under a change of variables (Jacobian transformation) | **C)** To compensate for Malmquist bias in massive galaxies | **D)** To align the mass scale with the Hubble parameter

Correct: B

Q9: In the bimodal distribution of galaxies, what is the term for the transitional region between the active 'Blue Cloud' and the passive 'Red Sequence'?

A) The starburst sequence | **B)** The Green Valley | **C)** The dust gap | **D)** The cosmic void

Correct: B

Q10: What empirical astrophysical scaling relation is represented by the Star Formation Main Sequence (SFMS)?

A) The relationship between stellar mass and central black hole mass | **B)** The correlation between a galaxy's stellar mass and its star formation rate | **C)** The relationship between orbital velocity and galactic radius | **D)** The ratio of gas mass to total dark matter mass

Correct: B

Q11: If a galaxy's Specific Star Formation Rate (sSFR) falls below 10^{-11} yr^{-1} , what does this indicate about its evolutionary state?

A) It is in an active starburst phase | **B)** It is undergoing a major gas-rich merger | **C)** It is quenched (passive/retired from forming stars) | **D)** It is collapsing into a supermassive black hole

Correct: C

Q12: What does the typical scatter of ~ 0.3 dex around the Star Formation Main Sequence (SFMS) represent physically?

A) Instrumental measurement noise in spectroscopic surveys | **B)** Stochastic variations in star formation histories from episodic accretion and feedback | **C)** Relativistic length contraction and lensing distortions | **D)** Gravitational redshift effects near the galactic center

Correct: B

Q13: What selection effect causes flux-limited astronomical catalogs to be dominated by intrinsically luminous objects at high redshifts?

A) Eddington bias | **B)** Lutz-Kelker bias | **C)** Malmquist Bias | **D)** Luminosity-distance degeneracy

Correct: C

Q14: Why does a young, star-forming blue galaxy exhibit a much lower mass-to-light ratio than an old red galaxy of the same stellar mass?

A) Blue galaxies are dominated by massive, hot O and B stars that emit high luminosity | **B)** Red galaxies contain significantly more dark matter in their cores | **C)** Blue galaxies are completely devoid of interstellar dust obscuration | **D)** Red galaxies are expanding away from the observer at a faster rate

Correct: A

Q15: In a galaxy survey with a spectroscopic limit of apparent magnitude $m = 23.5$, why is a galaxy with $m = 24.8$ excluded?

A) Its radial velocity exceeds the Hubble flow limit | **B)** It is below the detection threshold because it is too faint (higher apparent magnitude) | **C)** It is classified as a foreground stellar contaminant | **D)** Its light has been completely absorbed by cosmic dust

Correct: B

Q16: By definition, absolute magnitude is the apparent magnitude a celestial object would exhibit if observed from a standard distance of:

A) 1 Astronomical Unit (AU) | **B)** 1 Megaparsec (Mpc) | **C)** 10 parsecs | **D)** 1 Light-Year

Correct: C

Q17: Which stellar spectral class corresponds to the hottest, most massive main-sequence stars that terminate as core-collapse supernovae?

A) M-class (red dwarfs) | **B)** K-class (orange dwarfs) | **C)** A-class (white stars) | **D)** O-class (blue supergiants)

Correct: D

Q18: Which class of supernova is triggered by the thermonuclear runaway of a carbon-oxygen white dwarf exceeding the Chandrasekhar limit in a binary system?

A) Type II Supernova | **B)** Type Ia Supernova | **C)** Kilonova | **D)** Core-Collapse Supernova

Correct: B

Q19: Which event is triggered when a star with an initial mass $M > 8 M_{\text{sun}}$ forms an iron core that collapses under gravity?

A) Type II Supernova (Core-Collapse) | **B)** Type Ia Supernova | **C)** Helium Flash | **D)** Planetary Nebula ejection

Correct: A

Q20: What component of spiral galaxies is required to explain the flat rotation curves observed at large radii, where visible matter density is low?

A) A central supermassive black hole | **B)** A massive, extended Dark Matter Halo | **C)** Interstellar magnetic fields | **D)** Peculiar velocities from metric expansion

Correct: B

Q21: Using Hubble's Law ($v = H_0 \cdot d$) with a Hubble constant $H_0 = 70 \text{ km/s/Mpc}$, what is the recession velocity of a galaxy at a distance of 2 Megaparsecs?

A) 70 km/s | **B)** 140 km/s | **C)** 35 km/s | **D)** 280 km/s

Correct: B

Q22: What is the primary physical cause of cosmological redshift in the spectrum of distant galaxies?

A) Doppler shifts from galaxies moving physically through static space | **B)** Scattering of light by interstellar and intergalactic dust | **C)** The expansion of the space metric itself stretching light wavelengths during propagation | **D)** Gravitational time dilation near massive galaxy clusters

Correct: C

Q23: What is the term for the cosmological postulate that, on sufficiently large scales, the universe is homogeneous and isotropic?

A) The Cosmological Principle | **B)** The Copernican Principle | **C)** The Equivalence Principle | **D)** The Einstein Postulate

Correct: A

Q24: What is the expression for the Schwarzschild radius (R_s), which defines the event horizon of a spherically symmetric, non-rotating black hole?

A) $R_s = GM / c^2$ | **B)** $R_s = 2GM / c^2$ | **C)** $R_s = GM / (2c^2)$ | **D)** $R_s = \sqrt{GM / c}$

Correct: B

Q25: Which light element was synthesized in trace amounts during Big Bang Nucleosynthesis, in addition to hydrogen and helium?

A) Carbon | **B)** Iron | **C)** Lithium | **D)** Oxygen

Correct: C

Q26: Excluding local peculiar velocities, how does the comoving distance between two distant galaxies change as the universe expands?

A) It increases proportionally to the scale factor $a(t)$ | **B)** It remains constant by definition | **C)** It decreases due to local gravitational attraction | **D)** It increases exponentially due to dark energy

Correct: B

Q27: Why does stellar and supernova feedback suppress star formation in dwarf galaxies much more effectively than in massive galaxies?

A) Dwarf galaxies lack dark matter halos entirely | **B)** Dwarf galaxies have shallow potential wells and low escape velocities | **C)** Dwarf galaxies have higher star formation efficiency | **D)** Dwarf galaxies are closer to the cosmic web filaments

Correct: B

Q28: The Star Formation Main Sequence (SFMS) has a sub-linear slope ($b \sim 0.8$) in $\log \text{SFR}$ vs $\log M_{\text{star}}$. What does this indicate about galaxy growth efficiency?

A) Massive galaxies form stars more efficiently per unit mass | **B)** Low-mass galaxies form stars more efficiently per unit mass (higher sSFR) | **C)** Star formation rates are independent of stellar mass | **D)** Lower-mass galaxies are fully quenched

Correct: B

Q29: Two identical stars are observed, with Star B situated exactly 10 times further away than Star A. By how many magnitudes fainter will Star B appear?

A) 2.5 magnitudes fainter | **B)** 5.0 magnitudes fainter | **C)** 10.0 magnitudes fainter | **D)** 100.0 magnitudes fainter

Correct: B

Q30: Why did Big Bang Nucleosynthesis fail to synthesize carbon, leaving the early universe with only light elements?

A) The nuclear binding energy of carbon is unstable under any conditions | **B)** The universe cooled and diluted too rapidly to permit the triple-alpha helium fusion process | **C)** Dark matter particles absorbed the required free neutrons | **D)** Strong stellar winds from early stars dispersed the helium nuclei

Correct: B